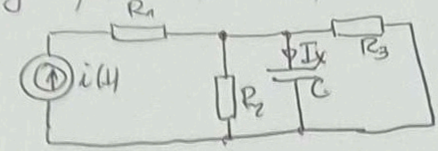
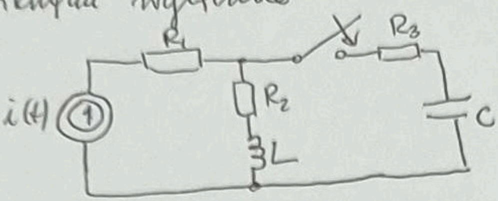
TOIS 24.06.25

1. Obliczyć prąd I_x stosując metodę Thevenina.



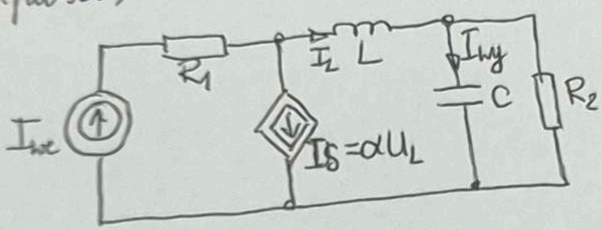
$R_1 = 1\Omega$
$R_2 = 2\Omega$
$R_3 = 3\Omega$
$\frac{1}{\omega C} = 2\Omega$
$i(t) = 2\sqrt{2}\sin(\omega t - 90^\circ)$

2. Określić $i_C(t)$, $u_C(t)$ w stanie nieustalonym po zamknięciu wyłącznika



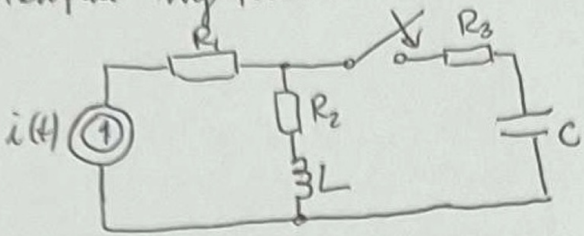
$R_1 = R_2 = R_3 = 2\Omega$
$L = 1H$
$C = 1F$
$i(t) = 10A$

3. Określić transmitancję $H(s) = \frac{I_{wy}(s)}{I_{we}(s)}$ w obwodzie, odpowiedź impulsową i charakterystyki częstotliwościowe.



$R_1 = 2\Omega$
$L = 1H$
$C = 1F$
$R_2 = 1\Omega$
$\alpha = 4$

2. Określić $i_C(t)$, $u_C(t)$ w stanie nieustalonym po zamknięciu wyłącznika



$R_1 = R_2 = R_3 = 2\Omega$
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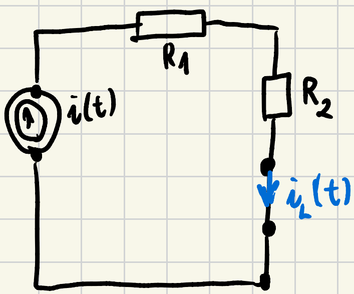
$C = 1F$
$i(t) = 10A$

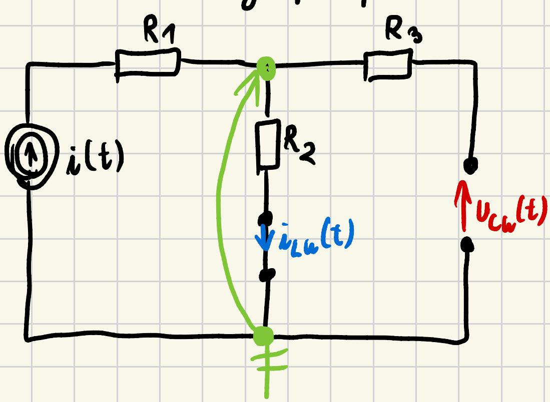
1° Warunki początkowe



$i_L(t) = 10A \qquad u_C(t) = 0V$

$i_L(0^-) = 10A \qquad u_C(0^-) = 0V$

2° Stan ustalony po przełączeniu



$i_{Lu}(t) = 10A$

$i_{Lu}(0^+) = 10A$

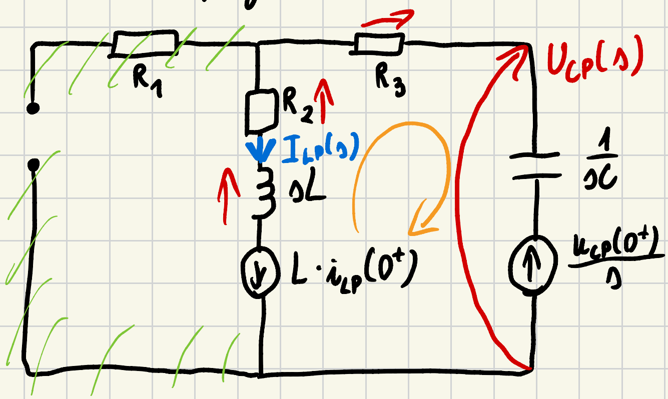
$u_{Cu}(t) \cdot \frac{1}{R_2} = I$

$u_{Cu}(t) = I \cdot R_2$

$u_{Cu}(t) = 10 \cdot 2 = 20\ V$

$u_{Cu}(0^+) = 20\ [V]$

3° Składowa przejściowa



$i_{cp}(t) = i_{Lp}(t)$

m NPK

$$I_{Lp}(s)\left[sL + R_2 + R_3 + \frac{1}{sC}\right] - L\cdot i_{Lp}(0^+) - \frac{u_{cp}(0^+)}{s} = 0$$

$$I_{Lp}(s)\left[sL + R_2 + R_3 + \frac{1}{sC}\right] = L\cdot i_{Lp}(0^+) + \frac{u_{cp}(0^+)}{s}$$

$$i_L(0^-) = i_{Lw}(0^+) + i_{Lp}(0^+)$$

$$i_{Lp}(0^+) = i_L(0^-) - i_{Lw}(0^+) = 10 - 10 = 0\ \text{A}$$

$$u_{cp}(0^+) = u_c(0^-) - u_{cw}(0^+) = 0 - 20 = -20\ \text{V}$$

$$I_{Lp}(s)\left[s + 4 + \frac{1}{s}\right] = 0 - \frac{20}{s} \quad /\cdot s$$

$$I_{Lp}(s)\left[s^2 + 4s + 1\right] = -20$$

$$I_{Lp}(s) = \frac{-20}{s^2+4s+1} \rightarrow$$

$s^2 + 4s + 1 = 0$

$\Delta = 16 - 4 = 12 \qquad \sqrt{\Delta} = 3{,}46$

$s_1 = \frac{-4-3{,}46}{2} = -3{,}73$

$s_2 = \frac{-4+3{,}46}{2} = -0{,}27$

$$I_{Lp}(s) = \frac{-20}{(s+3{,}73)(s+0{,}27)}$$

$$i_{Lp}(t) = \lim_{s\to -3{,}73}\left(\frac{-20}{s+0{,}27}\cdot e^{st}\right) + \lim_{s\to -0{,}27}\left(\frac{-20}{s+3{,}73}\cdot e^{st}\right) =$$

$$i_{Lp}(t) = \frac{-20}{-3,73+0,27} \cdot e^{-3,73t} - \frac{20}{-0,27+3,73} \cdot e^{-0,27t} =$$

$$= 5,78\, e^{-3,73t} - 5,78\, e^{-0,27t}$$

$$i_{Cp}(t) = i_{Lp}(t) = 5,78 e^{-3,73t} - 5,78 e^{-0,27t}$$

$$i_C(t) = 5,78(e^{-3,73t} - e^{-0,27t})$$

$$i_L(t) = \overbrace{10}^{i_{Lu}(t)} + \overbrace{5,78 e^{-3,73t} - 5,78 e^{-0,27t}}^{i_{Lp}(t)}$$

$$U_{Cp}(s) = I_{Lp}(s) \cdot \frac{1}{sC} + \frac{u_{Cp}(0^+)}{s}$$

$$U_{Cp}(s) = \frac{-20}{s^2+4s+1} \cdot \frac{1}{s} - \frac{20}{s}$$

$$U_{Cp}(s) = \frac{-20}{s(s+3,73)(s+0,27)} - \frac{20}{s}$$

$$u_{Cp}(t) = \lim_{s \to 0} \left(\frac{-20}{(s+3,73)(s+0,27)} \cdot e^{st} \right) + \lim_{s \to -3,73} \left(\frac{-20}{s(s+0,27)} \cdot e^{st} \right) +$$

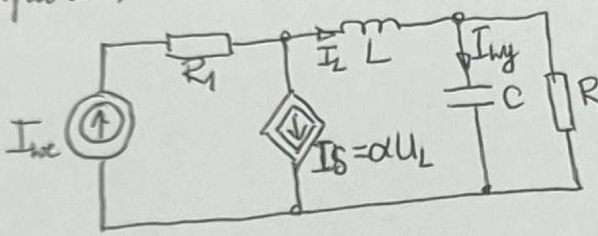
$$+ \lim_{s \to -0,27} \left(\frac{-20}{s(s+3,73)} \cdot e^{st} \right) - 20$$

$$u_{Cp}(t) = -19,86 - 1,55 e^{-3,73t} + 21,41 \cdot e^{-0,27t} - 20$$

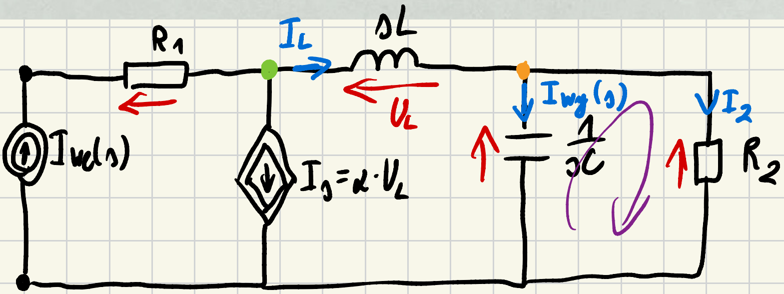
$$u_{Cp}(t) = -1,55 e^{-3,73t} + 21,41 e^{-0,27t} - 39,86$$

$$u_C(t) =$$

3. Określić transmitancję $H(s)=\frac{I_{wy}(s)}{I_{we}(s)}$ w obwodzie, odpowiedź impulsową i charakterystyki częstotliwościowe.



$R_1=2\Omega$
$L=1H$
$C=1F$
$R_2=1\Omega$
$\alpha=4$



$$I_{we}(s) = I_L + I_s = I_L + \alpha\cdot U_L = I_L + \alpha\cdot sL\cdot I_L = I_L + 4sI_L$$

$$I_{we}(s)=I_L(4s+1) \Rightarrow I_L = \frac{I_{we}(s)}{4s+1}$$

$$I_L = I_{wy}(s)+I_2$$

$$I_{wy}(s)\cdot\frac{1}{sC} = I_2\cdot R_2$$

$$I_{wy}(s)=I_2\cdot s \Rightarrow I_2=\frac{I_{wy}(s)}{s}$$

$$I_L = I_{wy}(s)\left[1+\frac{1}{s}\right]$$

$$I_{wy}(s)\left[1+\frac{1}{s}\right]=\frac{I_{WE}(s)}{4s+1}$$

$$I_{wy}(s)[s+1]=\frac{s}{4s+1}I_{WE}(s)$$

$$H(s) = \frac{s}{(s+1)(1+4s)}$$

Odp. impulsowa

$$y(t) = L^{-1}[H(s)] = \lim_{s \to -1}\left(\frac{s}{1+4s} e^{st}\right) + \lim_{s \to -\frac{1}{4}}\left(\frac{s}{s+1} e^{st}\right) =$$

$$= \frac{-1}{-3} e^{-t} + \frac{-\frac{1}{4}}{-\frac{1}{4}+1} e^{-\frac{1}{4}t} = \frac{1}{3} e^{-t} - \frac{1}{3} e^{-\frac{1}{4}t}$$

Ch. częstotliwościowe

$$H(s) = \frac{s}{(s+1)(1+4s)} = \frac{s}{s+4s^2+1+4s} = \frac{s}{4s^2+5s+1}$$

$$H(j\omega) = \frac{j\omega}{-4\omega^2+5j\omega+1} = \frac{0+j\omega}{1-4\omega^2+5j\omega}$$

$$|H(\omega)| = \frac{\sqrt{0^2+\omega^2}}{\sqrt{(1-4\omega^2)^2+(5\omega)^2}} = \frac{\omega}{\sqrt{(1-4\omega^2)^2+25\omega^2}}$$

$$\varphi(\omega) = \mathrm{arctg}\left(\frac{\omega}{0}\right) - \mathrm{arctg}\left(\frac{5\omega}{1-4\omega^2}\right)$$

$$H(s=j\omega) = \frac{A(\omega)+jB(\omega)}{C(\omega)+jD(\omega)} = H(\omega)$$

$$|H(\omega)| = \frac{\sqrt{A^2(\omega)+B^2(\omega)}}{\sqrt{C^2(\omega)+D^2(\omega)}}$$ → charakterystyka amplitudowa

$$\varphi(\omega) = \mathrm{arctg}\left(\frac{B(\omega)}{A(\omega)}\right) - \mathrm{arctg}\left(\frac{D(\omega)}{C(\omega)}\right)$$ → charakterystyka fazowa

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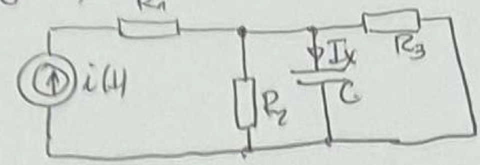

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$R_2 = 2\Omega$

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$\frac{1}{\omega C} = 2\Omega$

$i(t) = 2\sqrt{2}\sin(\omega t - 90^\circ)$

1° Wyznaczam Z_x

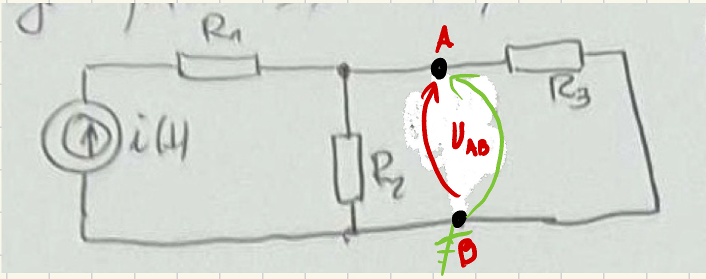
$$I = -j2$$

$$Z_x = Z_C = -j \cdot \frac{1}{\omega C} = -j2\ [\Omega]$$

2° Wyznaczam R_{AB}

$$R_{AB} = \frac{R_2 \cdot R_3}{R_2 + R_3} = \frac{2 \cdot 3}{2+3} = \frac{6}{5} = 1{,}2\ [\Omega]$$

3° Wyznaczam U_{AB}



$$I_x = \frac{U_{AB}}{R_x + R_{AB}}$$

$$I_x = \frac{-j2{,}4}{1{,}2 - j2}$$

$$I_x = 0{,}88 - j0{,}53$$

$$U_{AB}\left[\frac{1}{R_2} + \frac{1}{R_3}\right] = I$$

$$U_{AB} = I \cdot \frac{R_2 \cdot R_3}{R_2 + R_3} = -j2 \cdot 1{,}2 = -j2{,}4\ [V]$$

$$I_x =$$